

One Hot Finite State Machine

```
module one_hot_fsm1
```

```
input wire clk;
```

```
input wire rst;
```

```
input wire go;
```

```
input wire jump;
```

```
output reg y1;
```

```
;
```

```
reg [9:0] state, next;
```

```
localparam S0 = 10'b0000000001;
```

```
localparam S1 = 10'b0000000010;
```

```
localparam S2 = 10'b0000000100;
```

```
localparam S3 = 10'b0000001000;
```

```
localparam S4 = 10'b0000010000;
```

```
localparam S5 = 10'b0000100000;
```

```
localparam S6 = 10'b0001000000;
```

```
localparam S7 = 10'b0010000000;
```

```
localparam S8 = 10'b0100000000;
```

```
localparam S9 = 10'b1000000000;
```

```
always @(posedge clk or negedge rst)
```

```
begin
```

```
if (!rst)
```

```
state <= S0;
```

else

State ← next;

end

always @ (*) begin

next = state;

Y1 = 1'b0;

case (state).

S0: begin

Y1 = 1'b1;

if (go && jmp)

next = S1;

else

next = S0;

end

S1: begin

Y1 = 1'b0;

if (!jmp)

next = S2;

else

next = S3

end

S2: begin

Y1 = 1'b0;

next = S3;

~~if (j)~~

```
S3 : begin
    Y1 = 1'b1 ;
    if (jimp)
        next = S3 ;
    else
        next = S4 ;
end
```

```
S4 : begin
    Y1 = 1'b0 ;
    if (jimp)
        next = S7 ;
    else
        next = S5 ;
end
```

```
S5 : begin
    Y1 = 1'b0
    if (jimp)
        next = S3
    else
        next = S6
end
```

```
S6 : begin
    Y1 = 1'b0 ;
    if (jimp)
        next = S3 ;
    else
        next = S7 ;
end
```

```
S7 : begin
    Y1 = 1'b0 ;
    if (jimp)
        next = S8 ;
    else
```

```
        next = S8 ;
end
```

```
S8 : begin
    Y1 = 1'b0 ;
    if (jimp)
        next = S3 ;
    else
        next = S9 ;
end
```

S9 : begin

Y1 = 1'b1;

if (jimp)

next = S3;

else

next = S0;

end

default: begin

next = S0

Y1 = 1'b1;

end

endcase

end

endmodule